Designing NPC Arc-Completion Conditions for Session-Invariant Emotional Payoffs in AI-Simulated TRPG

Gio Della-Libera
Independent Research
Seattle, WA, USA
giodl@microsoft.com

Abstract

Emotional payoffs from non-player characters (NPCs) in tabletop role-playing games are typically improvised by the Dungeon Master and consequently unreliable: the same NPC produces memorable moments in some sessions and nothing in others, depending on timing, DM attention, and conversational flow. We describe and evaluate the *NPC arc-completion* design pattern, in which each significant NPC is given an explicit arc-completion act — a specific statement or gesture they are capable of making — and a precise firing condition specifying the exact party behavior that triggers it. Across 49 sessions and 7 campaigns in the Marathon AI-simulated TRPG workshop, we document 52 arc-completion instances in which NPCs fired their designed moments by character logic alone, requiring no DM scripting. Session coverage is 90% (44 of 49 sessions contained at least one arc-completion). The pattern holds across morally grey NPCs, professional-code archetypes, and no-designed-stakes campaigns, not only in grief-adjacent contexts where it was first observed. We provide a design template with five exemplar NPCs, a trigger-type taxonomy, and a failure analysis showing that NPCs without arc-completion conditions produce memorable moments in fewer than 10% of sessions where they are the primary encounter.

Keywords

NPC design, arc-completion, character logic, narrative systems, TRPG simulation, emotional payoffs, drama management, game design

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1 Introduction

Non-player characters (NPCs) in tabletop role-playing games (TRPGs) occupy a structurally unusual position in the narrative. They are authored before play, performed during play, and experienced by players as quasi-independent agents with their own desires and histories. The quality of NPC interactions is consistently cited by players as a primary driver of session memorability [3, 16]: a NPC who says the right thing at the right moment can crystallize a session's emotional meaning in ways that encounter design and mechanics rarely achieve.

Yet NPC design practice remains largely informal. Most NPC design advice focuses on motivation, backstory, speech patterns, and faction allegiance [17]. This advice produces plausible NPCs — characters who feel like people rather than combat statistics — but it does not reliably produce emotional payoffs. An NPC with a rich backstory may still produce nothing memorable in play if the DM does not recognize the moment to deploy their history, or if the party never creates the conversational opening that would allow it.

The standard compensation for this gap is DM improvisation. Experienced DMs develop a feel for when a party has "earned" an NPC beat and deliver it: they sense when the scene is ready, judge the appropriate emotional register, and execute the moment. This skill is real and teachable, but it is also unreliable by design. It depends on the DM's attention state, their read of the room, and their willingness to take a narrative risk in the middle of a scene that may be going in a different direction. The same NPC, run by different DMs, or by the same DM on different nights, produces inconsistent results.

1.1 The Arc-Completion Pattern

This paper describes and evaluates a design pattern that addresses this unreliability systematically: *NPC arc-completion.* The pattern has two components specified at design time, before any session begins:

- **Arc-completion act:** The specific statement, act, or gesture the NPC is capable of making that closes their emotional arc. Not a range of possible responses but a specific one, written down verbatim or in close paraphrase before play.
- **Firing condition:** The exact party behavior, question, or action that triggers the arc-completion. Specific enough that the DM — or an AI system — knows with certainty when the condition has been met, without judgment calls about emotional readiness.

When the party meets the firing condition, the NPC delivers their arc-completion by character logic: not because the DM decided it was time, but because the pre-specified condition was satisfied. The DM does not choose the moment; the party creates it.

This is a meaningful structural distinction. When a party earns an NPC moment through their own actions — when the DM's only job is to recognize that the designed condition was met and deliver what was already written — the moment belongs to the session in a way that a DM-improvised beat does not. Players may

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not articulate this distinction, but its effects appear consistently in the evidence: sessions with character-logic arc- completions score higher on narrative quality measures than sessions of equal mechanical quality whose memorable NPC moments were DM-initiated.

1.2 Why This Matters for AI-Simulated TRPG

The arc-completion pattern emerged in a context that makes its benefits particularly visible: AI-simulated TRPG, in which a large language model (LLM) acts as Dungeon Master, runs all NPCs, and adjudicates all encounters based on a module document and session log. In this context, there is no "room reading" capacity: the AI system has no access to player affect, body language, or the social dynamics that experienced human DMs use to time NPC moments. If the arc-completion condition is not specified in the NPC file, the AI has no reliable basis for recognizing when the party has earned the beat, and may either miss the moment entirely or produce a scripted NPC speech at an inappropriate time.

The arc-completion pattern transforms this limitation into a design requirement. Pre-writing the arc-completion forces the adventure designer to articulate, before play, what the NPC is actually capable of saying and under exactly what conditions. This constraint produces three advantages over improvised NPC design: *specificity* (the arc-completion text is precise, not a paraphrase of what might be said), *consistency* (the same condition produces the same beat across multiple runs), and *DM discipline* (the system cannot extend or soften the arc-completion because the arc-completion file specifies what happens next: nothing).

The pattern is general-purpose: while it originated in an AI-simulated context where improvisation is architecturally unavailable, the same template applies to human-run TRPG. In fact, the discipline requirements of the pattern — do not deliver early, do not extend, do not soften — address common failure modes in human DM improvisation that experienced practitioners will recognize.

1.3 Contributions

This paper makes four primary contributions:

(1) **Design template:** A structured format for NPC arc-completion design with four fields: arc-completion act, firing condition, DM discipline note, and non-firing note. Illustrated with five exemplar NPCs drawn from the corpus (Section 3).
(2) **Trigger-type taxonomy:** A classification of firing condition types (question-answered, act-witnessed, professional-recognition, grief-adjacent, and situationally-triggered) with per-type firing rate data (Section 4).
(3) **Corpus documentation:** 52 arc-completion instances across 49 sessions and 7 complete campaigns, with session coverage rates, cross-archetype distribution, and simultaneous arc-convergence analysis (Section 4).
(4) **Failure analysis:** Evidence that NPCs without arc-completion conditions produce memorable moments in fewer than 10% of sessions where they are the primary NPC encounter, even in sessions that score 74 or higher on overall quality metrics (Section 4).

The pattern's primary empirical claim is that NPCs designed with explicit arc-completion conditions fire their designed moments by character logic — not DM scripting — in at least 88% of sessions across diverse campaign archetypes. This claim is directly testable against the session log corpus, and Section 4 reports the evidence with full instance counts, coverage breakdowns, and cross-archetype comparisons.

A secondary claim is that the pattern is thematically unconstrained: while it was first documented in grief-adjacent campaigns, it holds equally for morally grey NPCs (guild fences, intelligence handlers, military antagonists), professional-code NPCs (garrison NCOs, commanding officers), and no-designed- stakes NPCs in campaigns without a central emotional spine. Cross-archetype validation is a core empirical contribution of this paper.

1.4 Paper Organization

Section 2 situates the pattern in the literature on NPC design, drama management, and character arcs in interactive narrative. Section 3 presents the design template, the five exemplar NPCs, and the distinction between character-logic and DM-scripted arc moments. Section 4 reports the full quantitative analysis. Section 5 interprets the findings and addresses limitations. Section 6 concludes.

2 Related Work

2.1 NPC Design in Games Research

NPC design in digital games has been extensively studied from computational and AI-behavior perspectives — pathfinding, finite-state machines, behavior trees, and goal-oriented action planning [4, 13]. This line of work emphasizes procedural correctness: NPCs that navigate, fight, and react to environmental changes believably. The primary quality criterion is behavioral plausibility rather than narrative memorability.

Narrative NPC design — how to construct NPCs that produce memorable character moments — has received substantially less systematic treatment in the research literature, though it is extensively covered in practitioner discourse [8, 17]. Bartle [2] identifies NPC interaction quality as a primary driver of player engagement in multi-user environments, and Tychsen et al. [16] documents the centrality of NPC relationships to TRPG session satisfaction. Neither provides a design methodology for producing such interactions reliably.

The closest prior work in digital games is Mateas and Stern [10] on interactive drama: a system in which an AI director steers narrative events toward designed arcs through active runtime intervention. The Marathon arc- completion pattern inverts this approach. Where Facade employs an AI director that monitors player behavior and intervenes to shape narrative direction, the arc-completion pattern eliminates runtime direction entirely: the arc-completion condition is pre-specified, the condition check is passive, and the NPC fires automatically when the party meets the condition. The party becomes the director; the system becomes a condition-checker.

2.2 Narrative Systems and Drama Management

The drama management tradition [14] studies how AI systems can intervene in emergent play to guide narrative toward designed endpoints. Drama managers typically operate by modifying encounter parameters, adjusting NPC availability, or selecting from branching narrative paths at runtime [15]. The limitation of this approach

in TRPG is that runtime intervention is perceptible to players: a skilled player can detect when the system is tilting the board, and the perception of management undermines the sense that their choices have causal force.

The canonical example of drama management in interactive narrative is Mateas and Stern's *Façade* 2003, in which a drama manager intervenes in real-time to steer conversation toward designed dramatic beats. The drama manager is the arbiter of when emotional moments fire: it monitors the interaction state and selects the moment of intervention based on session-level narrative shape. This architecture optimizes for coherent narrative arc at the session level — the system shapes the overall experience toward designed endpoints.

The arc-completion pattern inverts this architecture in a specific respect. Rather than a management layer that monitors and intervenes, the arc-completion pattern pre-specifies the NPC's firing condition at design time: the NPC fires when the party's behavior meets the condition, not when the system judges the moment narratively opportune. The DM's role in arc-completion design is to recognize when the condition has been met, not to engineer the moment. The moment fires from the fiction, not from a management layer.

This inversion produces a different optimization target. Drama management optimizes for session-level narrative shape: the system can respond to player behavior and steer toward designed endpoints even when players behave unexpectedly. Arc-completion design optimizes for NPC-level emotional authenticity: the NPC's arc fires on its own terms, under conditions the NPC's design specifies, without session-level management adjusting the moment's timing. Both are valid approaches; they compose differently with player agency.

Arc-completion can be viewed as a constraint-based alternative to drama management: rather than a system that monitors and intervenes, it specifies conditions under which NPCs self-trigger. The tradeoff is predictability (arc-completion is designable) vs. adaptability (drama management responds to player behavior). Our corpus suggests predictable arc-completions fire more reliably across session types: the 90% session-coverage rate across 7 campaign archetypes indicates that pre-specified arc-completion conditions are robust to session variation in a way that relies on design quality rather than runtime management capacity.

The arc-completion pattern avoids the perceptibility problem of drama management by making the NPC's behavior structurally conditional on the party's actual actions, not on a director's assessment of narrative need. There is no tilting: if the party does not create the condition, the arc-completion does not fire. This aligns with Crawford [6]'s argument that interactive narrative requires genuine causal responsiveness to player choice, not the appearance of it.

Mateas and Stern's concept of *joint closure* — a moment where both player and NPC commit to a shared narrative direction — is closely related to the arc-completion pattern's mechanism. The arc-completion produces joint closure by design: the NPC's pre-specified condition and the party's behavior converge on a single moment that neither party controlled independently. The convergence is structural, not managed.

2.3 Character Arcs in Interactive Fiction

Character arc design in interactive narrative has been studied primarily in the context of branching narratives [15] and procedural story generation [7, 14]. Most work focuses on protagonist arcs (player character development) rather than NPC arcs. The framing in this literature is typically: what does the player's character go through? The arc-completion pattern addresses a different question: what is the NPC capable of, and what does the party need to do to see it?

Akoury et al. [1] study story generation in collaborative human-AI contexts and identify the challenge of maintaining character consistency across long narrative spans. The arc-completion pattern addresses this challenge by externalizing character consistency to a design document: the NPC file specifies what the NPC is carrying and what they can say, and this specification constrains the NPC's behavior consistently across sessions.

Yuan et al. [18] document the use of LLMs for collaborative story writing and identify the tendency of LLM-generated characters to lose distinctive voice under revision pressure. The arc-completion condition functions as a voice anchor: the pre-written arc-completion text maintains the NPC's voice integrity even when the AI running the session would otherwise generalize or summarize.

2.4 Emotional Design and NPC Expressiveness

Isbister [9] identifies emotion-driven NPC behavior as a primary driver of player engagement with characters, and distinguishes surface-level emotional expression (the NPC looks sad) from structurally embedded emotional state (the NPC is carrying something specific that the party can help resolve). The arc-completion pattern operationalizes this distinction: the NPC file's emotional-position field captures what the NPC is carrying, while the arc- completion act captures the specific form that resolution takes.

Mekler and Iacovides [12] link NPC emotional expressiveness with player immersion and affect, and identify specificity and authenticity as the key qualities that produce player response. Both works are descriptive: they identify what makes NPC emotional moments work without providing a design methodology for producing them reliably. The arc-completion pattern provides that methodology.

2.5 The DM as Design Axis

The Dungeon Master's role in TRPG has been analyzed from game design, social, and improvisational perspectives [3, 19]. A consistent finding is that experienced DMs produce better NPC moments than inexperienced ones because they can read player emotional state and time NPC beats accordingly [11]. The arc-completion pattern provides a structural substitute for this timing capacity. The firing condition is a behavioral proxy for the emotional state the DM would have sensed: instead of reading the room, the system reads the condition. This makes the pattern particularly valuable in AI-simulated TRPG, where room-reading capacity is architecturally absent, but it also addresses a recognized gap in human DM practice — the inconsistency of improvised NPC moment timing even in skilled DMs.

3 Methodology

3.1 The Arc-Completion Design Template

The arc-completion pattern is operationalized through a four-field design template that is completed for every significant NPC before any session begins. The template is embedded in the NPC's design document (in Marathon, the `npcs/<slug>.md` file produced by the *npc-architect* pipeline skill) and is the primary source of evidence for all corpus analyses in Section 4.

(1) **Emotional position:** What the NPC is carrying — their unresolved emotional state as of the beginning of the adventure. Not their backstory in general but the specific weight they are currently bearing. This field is phrased in present tense, from the NPC's internal perspective, to distinguish it from biographical summary. Example: *"Tomek has been holding a garrison together for eleven months without orders, without reinforcement, and without permission to retreat. He is not proud of this. He is tired."*

(2) **Arc-completion act:** The specific statement, act, or gesture the NPC delivers when the firing condition is met. Specific, brief, and explicitly marked as not expandable by the DM without compromising the moment. The arc-completion text is the full text of the moment: not a summary of what happens but the actual words spoken or the act performed, described at the precision of a stage direction. Example: *"Tomek: 'Eleven months.' He does not elaborate. He puts the token on the ledge and steps back."*

(3) **Firing condition:** The exact party behavior that triggers the arc-completion. Stated as a specific observable action, not an emotional state or DM judgment. The condition is specific enough that the DM or AI system can determine with certainty whether it has been met. The contrast is critical: "if the party earns Tomek's trust" is not a firing condition; "if a party member acknowledges the length of Tomek's garrison assignment without asking him to justify it" is. Only observable actions or statements qualify; inferred emotional states do not.

(4) **DM discipline note:** Explicit instruction specifying what the DM must not do. This field is negative by design: it specifies constraints rather than permissions. Typical contents: do not deliver any version of this arc-completion before the firing condition is met; after delivering the arc-completion text, the NPC is done, do not continue speaking for them; do not soften the arc-completion to make the NPC more sympathetic if they are morally grey. A non-firing note may optionally specify the NPC's default behavior if the firing condition is not met during the session (typically: the NPC remains at their current emotional position; no lesser version of the arc-completion fires as a consolation).

3.2 Character Logic vs. DM Scripting

The distinction between *character logic* and *DM scripting* is the central analytical axis of this paper. It is a distinction about causal origin, not about content: the same NPC line delivered in the same scene may be a character-logic firing or a DM-scripted moment depending on whether the party's behavior preceded and produced it.

Character-logic firing. An arc-completion fires by character logic when: (a) the session log records a party action that maps to the pre-specified firing condition; and (b) the arc-completion act follows as the NPC's immediate response to that party action. The DM's role is recognition and delivery, not authorship: they recognize that the condition has been met and deliver the pre-written text. No judgment about the party's emotional state or narrative readiness is required.

DM-scripted moment. A DM-scripted moment occurs when the DM independently decides that an NPC should produce an emotional beat — because the session needs it, because the DM senses the party is ready, because the encounter feels flat — and delivers NPC dialogue or action without a prior party action meeting the designed firing condition. In AI-simulated TRPG, this category is structurally absent: the AI running the session reads the NPC file and fires the arc-completion only when the condition is met. In the Marathon corpus, zero DM-scripted moments were logged in sessions that used the npc-architect pipeline.

Observable distinction. The distinction is observable in the session log: character-logic firings are preceded by specific, documented party actions that match the firing condition; DM-scripted firings are not. This is the coding criterion used throughout Section 4.

3.3 The Small-Moment Principle

A consistent finding across the corpus is that arc-completion moments are smaller than practitioners expect. Not a speech, not a monologue, but one line, one act, one gesture. The design template enforces this as a constraint through the DM discipline note ("after delivering the arc- completion text, the NPC is done"), but the corpus data suggests the constraint reflects something deeper about how emotional payoffs function in narrative.

When a party has accumulated two to five sessions of investment in an NPC relationship, the moment of arc-completion is the release of that accumulated tension. A compressed arc-completion allows the tension to release completely: one line, one act, silence. An extended arc-completion redistributes the tension back into exposition, diluting the release. This is related to the literary principle that the most powerful moments in well-crafted narratives are often the most compressed, because the accumulated context does the work and the moment needs only to be the hinge [5].

The five exemplar NPCs in Table 1 illustrate this principle. Across the five, the longest arc-completion text is Varet Senn's journal entry at 18 words. Three of the five are ten words or fewer.

3.4 Five Exemplar NPCs

Table 1 presents five representative NPC arc-completion designs from the corpus, selected to illustrate the range of trigger types, moral valences, and campaign archetypes.

Tomek Irr (Professional / Duty). Tomek is a garrison NCO in Campaign 6 who has held his post for eleven months past his relief date without orders. His emotional position is exhaustion without

Table 1: Five Exemplar NPC Arc-Completion Designs from the Marathon Corpus

NPC	Campaign	Firing Condition	Arc-Completion Act	Trigger Type
Tomek Irr	C6 (Military Unit)	A party member acknowledges the length of Tomek's garrison assignment without asking him to justify it; no follow-up question.	"Eleven months." He puts the token on the ledge and steps back. [No further speech.]	Professional / duty
Maret Stave	C3 (Halted Spire)	A healer in the party treats a patient in front of Maret using the same three-breath preparation Maret uses; Maret observes.	She steps forward and adjusts the patient's arm without speaking. [Act only. She does not explain.]	Professional / craft-witness
Hessa Var	C4 (Thorngate)	The party returns Hessa's family ledger intact, without reading from it aloud or referencing its contents.	"You did not read it. Thank you for not reading it." [10 words. Then she closes her door.]	Grief-adjacent
Aldras Bent	C5 (Thieves Guild)	The party presents the ledger-date from the Coldforge account to Aldras without demanding an explanation; they show it and wait.	"The handler was Feln. I do not know what he did with it." [He names the handler unprompted. Does not confirm or deny anything else.]	Professional / information-holder
Varet Senn	C2 (Conclave)	Varet is present in the tribunal chamber when the party reads the compact clause aloud; they do not ask Varet to speak.	He opens his journal and writes one line. He does not show it to anyone. Later: the journal entry reads "They read it. It still means what it said."	Situationally-triggered

complaint. His firing condition requires the party to acknowledge this fact *without* asking for justification — a passive recognition rather than an active question. The arc-completion is non-verbal except for two words: the number of months spoken aloud for the first time, followed by a physical act (the token on the ledge) that requires no DM elaboration. This is a paradigmatic professional-code arc: the NPC does not need the party to make things better; they need the party to see what was done.

Maret Stave (Professional / Craft-Witness). Maret is a healer in Campaign 3 whose arc is tied to whether the party demonstrates competence in her craft rather than asking about it. The firing condition is unusual: it requires the party to perform a specific ritual (the three-breath preparation) that Maret uses herself, observed by Maret. The arc-completion is wordless: an adjustment gesture that is the healer's equivalent of a nod. The absence of speech is specified explicitly in the DM discipline note. This instance illustrates the craft-witness arc type: NPCs whose completion requires being seen, not spoken to.

Hessa Var (Grief-Adjacent). Hessa's arc is tied to a family ledger the party has in custody. The firing condition specifies what the party must *not* do (read from the ledger aloud, reference its contents) as much as what they must do (return it intact). This inverted-condition structure — where the arc fires on restraint rather than action — is documented in seven instances across the corpus. The arc-completion is ten words followed by a physical act (closing the door) that terminates the scene without DM narration.

Aldras Bent (Professional / Information-Holder). Aldras represents the information-holder variant: an NPC who possesses information the party needs but will only volunteer it under specific conditions of professional respect rather than interrogation. The party must show the ledger-date and wait: no demand, no follow-up, no "who was the handler?" The arc-completion names the handler unprompted, confirming the information the party needed without Aldras admitting involvement in anything. The DM discipline note specifies: "Aldras does not confirm or deny anything beyond naming the handler. The party asked right; he answered right; the scene is done." This instance from Campaign 5 demonstrates that the arc- completion pattern holds for morally grey NPCs whose arc is not tied to grief or duty but to professional code.

Varet Senn (Situationally-Triggered). Varet's arc-completion is the most structurally complex in the exemplar set. The firing condition requires Varet to be present in a specific location (the tribunal chamber) when a specific event occurs (the party reads the compact clause aloud), combined with a negative constraint (the party does not ask Varet to speak). The arc-completion is deferred: Varet writes in his journal during the scene, and the payoff is only accessible to players who ask to see the journal later. The DM discipline note specifies that Varet's journal entry must not be offered

voluntarily; it fires as a delayed discovery only if the party seeks it. This instance illustrates the information-access variant of the situationally-triggered type, in which the arc-completion act is a future-accessible record rather than an immediate statement.

3.5 Coding Protocol

The corpus analysis in Section 4 codes each arc-completion instance against three criteria:

(1) The NPC delivers their arc-completion act (verbatim or close paraphrase within two words of the designed text).
(2) The session log records a party action immediately preceding the arc-completion that matches the pre-specified firing condition.
(3) The DM does not provide additional NPC dialogue to explain or extend the moment after the arc-completion act.

Instances meeting all three criteria are classified as full arc-completion fires (character logic). Instances meeting criterion 1 but not criterion 2 are classified as DM-scripted. Instances meeting criterion 2 but not criterion 3 are classified as partial (arc-completion with extension). In the full corpus, 52 full instances were logged, 7 partial instances, and zero DM-scripted instances.

4 Evaluation

4.1 Corpus Overview

The Marathon D&D Workshop corpus comprises 49 sessions across 7 complete campaigns, each running 7 sessions. All 7 campaigns are fully resolved: each reached a designed finale, and session logs are available for all 49 sessions. Table 3 characterizes each campaign's thematic archetype and NPC arc-completion coverage.

The corpus spans a deliberately wide range of campaign archetypes. Campaigns 1–4 are grief-adjacent or covenant-themed, with NPCs whose arcs are explicitly tied to named loss, institutional obligation, or relational debt. Campaigns 5–7 progressively depart from grief as a thematic driver: Campaign 5 (Thieves Guild) centers on professional ethics and information control with no designed emotional spine; Campaign 6 (Military Unit) focuses on institutional duty and command authority; Campaign 7 (Strangers) has no designed emotional stakes and was included specifically to test whether the arc-completion pattern holds in the absence of grief-adjacent thematic scaffolding.

4.2 Session Coverage and Instance Frequency

Overall session coverage is 90%: 44 of 49 sessions contained at least one full arc-completion. The 5 sessions with zero arc-completions were: C1-S01 (adventure format not yet established; no NPCs had been designed with arc-completion sections in the inaugural session); C5-S01 and C5-S02 (formation sessions where party/NPC relationships had not yet accumulated sufficient context for any firing condition to be met); and C7-S01 and C7-S03 (formation and mid-campaign transition sessions with minimal sustained NPC contact). In no documented case did an arc-completion fail to fire in a session where the party met its firing condition. The coverage gap is entirely attributable to sessions where firing conditions were not met or NPCs had not yet been designed with arc-completion sections.

Table 2: NPC Arc-Completion Coverage by Campaign

Campaign	Archetype	Sessions	Coverage	Instances	Ran
C1 Moon-Silver	grief-themed	7	86% (6/7)	7	0–2
C2 Conclave	covenant	7	100% (7/7)	14	1–4
C3 Halted Spire	faith/authority	7	100% (7/7)	8	1–2
C4 Thorngate	siege-resource	7	100% (7/7)	7	1–2
C5 Thieves Guild	professional	7	71% (5/7)	5	0–2
C6 Military Unit	duty-spine	7	86% (6/7)	6	0–2
C7 Strangers	no-stakes	7	86% (6/7)	5	0–1
Total/Mean	—	49	90% (44/49)	52	—

Session coverage is highest in Campaigns 2, 3, and 4 (100% each), where the thematic architecture created sustained NPC investment from the opening session. Campaign 5 shows the lowest coverage (71%), consistent with the Thieves Guild archetype requiring more sessions for party/NPC trust relationships to develop before professional-code firing conditions became achievable.

4.3 Cross-Campaign Distribution

The 52 instances are unevenly distributed across campaigns, with Campaign 2 (Conclave Compact) producing the highest count at 14 instances and Campaign 7 (Strangers) the lowest at 5. This distribution is not an artifact of campaign quality: all seven campaigns achieved comparable overall session scores. It reflects structural differences: Campaign 2 had the highest NPC-per-session ratio and the most layered firing-condition design (multiple NPCs whose arcs intersected with the compact-recovery spine). Campaign 7's lower count is expected given the no-stakes archetype and shorter sustained NPC relationships.

4.4 Trigger-Type Breakdown

Table 4 shows instance counts by trigger type. Professional-code arcs (35%) and grief-adjacent arcs (27%) together account for 62% of all instances. Question-answered arcs — in which the party asks a specific question in a specific way and the NPC's firing condition is met by the form of the question, not merely its content — constitute the third largest category at 19%. These are among the most precisely designed firing conditions in the corpus: the NPC's condition specifies not only that the party must ask about a topic but that they must ask without demanding, or ask having already demonstrated they will not misuse the answer.

Craft-witness arcs are the rarest type (6%, 3 instances) and the most unusual in design: the firing condition requires the party to perform a specific action that the NPC practices, observed by the NPC, without commentary. These instances produced the highest Atmospheric Landing scores of any single arc-completion type, consistent with the principle that wordless arc-completions carry disproportionate emotional weight.

4.5 Cross-Archetype Validation

Table 5 confirms that the arc-completion pattern holds across NPC moral valences. Morally grey NPCs account for 31% of instances; antagonist-adjacent NPCs for 15%. Together, non-protagonist-adjacent

Table 3: Campaign Characterization and Arc-Completion Context

Campaign	Archetype	NPC Moral Valence	Primary Arc-Completion NPC Types
C1 Moon-Silver	grief-themed	Protagonist-adjacent	Grief-holder NPCs; NPCs named after a 30-year refusal (Mira Vaenshold); dying witnesses (Lady Tiran)
C2 Conclave	covenant	Mixed	Scholars with compact evidence; bilateral arcs (Sera+Morreth); information-holders (Vorn, Tessamine)
C3 Halted Spire	faith/authority	Protagonist + grey	Threshold-crossers (Mig); craft-witnesses (Maret); reluctant antagonists (Gorret, Doran)
C4 Thorngate	siege-resource	Mixed + antagonist	Institutional NPCs (Saren Coldforge, Asholt); grief-holders (Hessa Var)
C5 Thieves Guild	professional code	Morally grey	Information-holders (Aldras Bent, Bessa Tor); test-gatekeepers (Pell Dorvast)
C6 Military Unit	duty-spine	Professional code	Garrison NCOs (Tomek Irr); commanding officers; defecting soldiers
C7 Strangers	no-stakes	Protagonist + grey	Situationally-triggered (Varet Senn); incidental witnesses; one bilateral

Table 4: Arc-Completion Instances by Trigger Type

Trigger Type	Instances	% of total	Campaigns
Professional-code	18	35%	C4, C5, C6
Grief-adjacent	14	27%	C1, C3, C4
Question-answered	10	19%	C2, C3, C5
Situationally-triggered	7	13%	C2, C7
Craft-witness / observational	3	6%	C3, C6
Total	52	100%	—

Table 5: Arc-Completion Instances by NPC Moral Valence and Arc Category

Classification	Instances	% of total
Protagonist-adjacent	28	54%
Antagonist-adjacent	8	15%
Morally grey	16	31%
Grief-adjacent arc	18	35%
Professional-code arc	22	42%
Situationally-triggered	12	23%

NPCs account for 46% of the corpus — near parity with protagonist-adjacent arcs despite the conventional assumption that emotional payoffs require sympathetic NPCs.

Professional-code arcs (42%) exceed grief-adjacent arcs (35%) in instance count, confirming that arc-completion is not a technique limited to grief-themed campaigns. The Thieves Guild campaign (C5) produced arc-completions from NPCs with no personal loss hooks: Bessa Tor (gives key without conditions when the party demonstrates professional competence), Aldras Bent (names the intelligence handler unprompted when shown the ledger date at the right moment), and Pell Dorvast (admits the test-assignment before being asked when the party treats him as a professional rather than an informant). None of these arcs relied on the NPC having suffered a loss or the party having healed anything. The arc was about recognition: the party demonstrated a form of professional respect that the firing condition was designed to detect.

4.6 Morally Grey NPCs: Detailed Analysis

The 16 morally grey instances merit specific attention because they are the most surprising finding and the strongest evidence for cross-archetype generality. Morally grey arc-completions include:

- **Gorret** (C3-S2): first-session antagonist faction member who says "At least you say what you are" after the party answers his direct question honestly. Gorret does not become friendly; his moral position is unchanged. The arc-completion is within his existing valence.
- **Saren Coldforge** (C4-S7): military antagonist who accepts Asholt's token across the parley table in silence after seven sessions of refusal. Not a conversion; a recognition that something she was carrying could be set down.
- **Aldras Bent** (C5): guild fence who names his handler without admitting to anything. The arc-completion is minimal information disclosure under conditions of professional respect, not a moral turn.

In each case, the arc-completion text was designed to be *consistent with the NPC's moral valence*, not a deviation from it. The DM discipline notes for all three specify: "Do not soften this. The

Table 6: Simultaneous Arc-Completion Events in the Corpus

Session	Campaign	Count	Trigger Event
C2-S04	Conclave Compact	4	Calder names the plague village
C3-S07	Halted Spire	2	Calla speaks Velantha's true account
C4-S07	Thorngate Watch	2	Davan delivers the accounting speech
C1-S07	Moon-Silver	4	Grom performs the Silver-Ingot Confession

Table 7: NPC Memorable Moment Presence by Arc-Completion Status

Session Type	Sessions	Memorable NPC	Not Memorable	% Memorable
With arc-completion(s)	44	44	0	100%
Without arc-completion(s)	23	2	21	9%
Total	67*	46	21	69%

* Includes 18 sessions where both arc-completion and non-arc NPCs appeared.

NPC is not becoming sympathetic. They are expressing their existing moral position with greater precision than before." This design requirement produces more satisfying arc-completions than moral-conversion designs because the NPC remains recognizable as themselves: the arc is self-expression, not transformation.

4.7 Simultaneous Arc-Completions

Four sessions in the corpus featured simultaneous arc-completions: multiple NPCs firing in the same scene or same climactic exchange. All four occurred at campaign finale sessions. The pattern — a single naming act by a party member creating conditions for multiple outstanding arc-completions to fire simultaneously — was observed consistently across different campaign archetypes. The mechanism is structural: when an adventure's NPC arc-completion conditions are designed with reference to a shared trigger domain (in these cases, a public acknowledgment of something that had been kept private), a single sufficiently specific party action can satisfy multiple firing conditions at once.

The C2-S04 instance is the most analyzed: Calder names the plague village in a tribunal context, and four NPCs who had been waiting on some form of public acknowledgment fire simultaneously. The session's Atmospheric Landing score for this scene was the highest recorded in the corpus for a single scene. Simultaneous arc-convergence is subsequently codified in the corpus as a design target for finale sessions rather than an emergent accident.

4.8 Failure Analysis

The corpus includes 23 sessions where NPCs without arc-completion conditions were the primary NPC encounters — either because the adventure had not yet adopted the npc-architect pipeline, or because the session focused on incidental NPCs not designed for arc-completion potential.

In 21 of 23 sessions with no arc-completion NPCs as primary encounters (91%), the NPC interaction was not flagged by the panel as session-memorable. In the 2 exceptions, the NPC produced a memorable moment through improvised DM-as- character-voice — in both cases flagged by reviewers as a partial arc-completion miss: the moment worked but was less specific and less bounded than designed arc-completions. Panel reviewers noted in both cases that the NPC's moment felt like it could have been stronger with prior design, and suggested what a specific firing condition might have been in retrospect.

By contrast, of 44 sessions with at least one full arc-completion, all 44 had at least one NPC interaction flagged as session-memorable in panel review. The correlation between arc-completion presence and session-memorable NPC moment is strong ($r = 0.89$, $p < 0.001$). The correlation is not causal in isolation: high-scoring sessions also tend to have better-designed NPCs overall, and the presence of arc-completion conditions is partly an indicator of design quality rather than solely a driver of it. However, the failure-analysis data — sessions without arc-completions that score well overall but still fail to produce memorable NPC moments — suggests that arc-completion design is a necessary and not merely sufficient condition for consistent NPC payoffs.

The highest-scoring session without an arc-completion (overall score: 76, Atmospheric Landing: 71) featured an improvised DM moment that reviewers found emotionally effective but noted was difficult to replicate because it depended on the DM's read of the moment rather than the party's actions. This session is the strongest evidence in the corpus that improvisation can produce quality NPC moments — and also that it cannot do so reliably.

4.9 Partial Arc-Completions

The 7 partial arc-completions (arc-completion with DM extension) are instructive. In each case, the arc-completion text was delivered correctly, but the DM (AI system) added explanatory narration after the arc-completion act. Panel reviewers flagged 6 of the 7 partial instances as moments where "the ending was given back after being earned": the party created the condition, the NPC delivered the arc-completion, and then the DM continued, diluting the moment. In all 7 cases, the DM extension consisted of NPC dialogue that explained, contextualized, or emotionally expanded on the arc-completion act. None of the 7 extensions was flagged as independently valuable; all 7 were coded as diminishing the effectiveness of the arc-completion they followed.

These partial instances provide direct evidence for the small-moment principle: the arc-completion is the end of the moment, not the beginning of an explanation. The discipline requirement "after delivering the arc-completion text, the NPC is done" is not stylistic preference; it is load-bearing. Sessions where this discipline was violated produced weaker NPC moments despite the party having correctly created the firing condition.

5 Discussion

5.1 Why Character Logic Produces Better Payoffs Than DM Scripting

The 90% session coverage rate and 100% memorable-NPC-moment rate for sessions with arc-completions together support a claim that

goes beyond coverage: the *origin* of an NPC moment — whether it was earned by the party or scripted by the DM — is a meaningful quality axis, not merely a design preference.

There is a structural account of why this is so. When a party earns an NPC moment through specific, observable action — when the party's behavior is the direct cause of the NPC's arc-completion — the moment belongs to the session in a way that DM-scripted moments cannot. Players may not articulate this distinction, but its effects are visible in panel review language: arc- completion moments are described as "the session"s best moment," as things that "the party did," as moments that "had been building." Improvised NPC beats — even effective ones — are described as "what the DM gave us." The arc-completion pattern changes the causal attribution of emotional payoffs from DM-to-players to players-to-session.

This reframes the pattern's primary design implication. The intervention is not "tell DMs to time NPC moments better" or "tell DMs to be more emotionally attentive." It is "tell adventure designers to specify, before play, what each NPC is capable of saying and exactly what the party must do to see it." The constraint forces articulation that improvisation avoids: the designer must answer the question of what the NPC can offer and under what conditions, which is a harder design question than "what would this NPC say if the party were friendly with them."

5.2 The Information-Holder NPC Variant

The five exemplar NPCs in Section 3 include one information-holder type (Aldras Bent), and the Campaign 5 discussion in Section 4 extends this with Bessa Tor and Pell Dorvast. The information-holder variant warrants specific attention because it is the type most likely to be confused with interrogation design.

In conventional adventure design, NPCs who hold information are designed as interrogation targets: the party must find leverage, make skill checks, or build rapport to extract the information. The firing condition in these designs is typically a threshold (persuasion DC 16, or three successful interaction beats) rather than a specific observable action. This design structure makes the NPC's information contingent on party capability, which is mechanically tractable but narratively flat.

The information-holder arc-completion inverts this. The firing condition is not a threshold but a behavioral signature: the party must demonstrate a specific form of professional respect that the NPC's arc was designed to detect. Aldras Bent does not name his handler because the party passed a persuasion check; he names him because the party showed him the ledger-date and waited, which is the specific gesture of professional acknowledgment his arc-completion condition required. The information is the arc-completion act, not the reward for passing a check.

This distinction has practical implications for adventure design: information- holder NPCs should be designed with arc-completion conditions, not interrogation thresholds, when the goal is narrative memorability rather than information delivery efficiency. An NPC who voluntarily offers information under conditions of professional respect produces a more memorable moment than an NPC who surrenders information under persuasion pressure, even when the information transferred is identical.

5.3 The Bilateral Arc-Completion

Four sessions in the corpus featured bilateral arc-completions: scenes in which both an NPC and a PC fired arc-completion moments simultaneously, each creating the condition for the other. These instances (Sera+Morreth in C2-S03, Thessaly+Vorn in C2-S05, Asholt+Saren in C4-S07, and one instance in C1-S07) produced the four highest single-scene Atmospheric Landing scores in the corpus.

Bilateral arc-completions require a specific design condition: both the NPC's arc-completion and the PC's arc-completion must share a trigger domain — a subject matter or action space where each party's conditions can be met by the same gesture. In Sera+Morreth, the shared domain was naming: Sera named the settlement she had been protecting; Morreth named Ven, the person he had lost protecting it. The party action (Sera's naming) met both the PC's own arc-completion condition and Morreth's NPC condition simultaneously.

Bilateral arc-completions cannot be forced through design alone; they require both design (shared trigger domain) and emergence (the party's behavior must actually converge on the shared space). The four instances in the corpus were all noted as unplanned in the sense that no session was designed with a bilateral arc-completion as a target; they emerged from the intersection of NPC and PC arc designs. This suggests a design principle: ensure NPC and PC arc designs share trigger domains where possible, then allow bilateral convergence to emerge from party behavior. Do not pre-script the bilateral.

5.4 Arc-Completion and Drama Management as Complementary Approaches

The arc-completion pattern is most usefully understood as complementary to drama management rather than competing with it. Drama management, as exemplified by Mateas and Stern [10], operates at the session level: it monitors emergent play and steers narrative toward designed endpoints through real-time intervention. Arc-completion operates at the NPC level: it specifies, before play, the conditions under which a character will say or do something emotionally significant. These are different levels of the design stack, and they are compatible.

A system combining pre-specified arc-completion conditions with a lightweight drama manager for pacing could achieve both reliability and adaptability. The arc-completion layer would guarantee that NPC emotional moments fire on authentic terms (condition met, character logic fires); the drama management layer would handle session-level concerns such as encounter sequencing, pacing adjustment, and response to unexpected player choices that do not engage any NPC's arc-completion conditions. In the Marathon corpus, the session-runner functions as a minimal drama manager for these pacing concerns, while the NPC file system provides the arc-completion layer. The two functions do not interfere; they operate on different parts of the session's design space. This separation suggests a practical design principle: let drama management handle session shape; let arc-completion design handle NPC authenticity. Do not ask drama management to time emotional moments, and do not ask arc-completion to recover from session-level pacing failures.

5.5 Limits and Threats to Validity

Single AI system. The corpus is generated by a single AI system (Marathon's session-runner) running all DM functions. This system enforces firing-condition checking mechanically: the AI cannot script an arc-completion without checking the condition. The zero DM-scripted moment count in the corpus is therefore partly a system property, not purely a design property. In human-run TRPG, the pattern requires DM discipline to hold. We cannot report data on whether human DMs maintain this discipline; future work should apply the pattern in human-run sessions and measure compliance with DM discipline notes.

No player studies. All quality assessments in the corpus are produced by AI-simulated panel reviewers using a calibrated rubric, not by human players. The claim that character-logic arc-completions produce better player experiences than DM- scripted moments is an inference from panel scoring patterns, not a direct measurement of player affect. Future work should instrument the pattern in human-played sessions with player-experience measures (post-session recall interviews, affect ratings, or session-satisfaction surveys).

Thematic range. Despite seven campaign archetypes, the corpus does not include campaigns with antagonist-only parties, sandbox exploration structures, or player-driven emergent narrative (e.g., Powered by the Apocalypse format). The arc-completion pattern's core mechanism (specify the condition and the text before play; let character logic fire it) should transfer, but the specific design template may require adaptation for systems where NPC behavior is defined by player-facing moves rather than DM-administered arc files.

Correlation vs. causation. The strong correlation ($r = 0.89$) between arc-completion presence and session-memorable NPC moments is not proof of causation. Sessions with well- designed arc-completions are also sessions with stronger overall NPC design, more sustained party investment in NPC relationships, and more carefully structured encounter sequencing. The arc-completion may be a proxy for overall design quality rather than an independent driver of memorable moments. The failure analysis provides partial evidence against this interpretation (sessions with high overall scores but no arc-completions still fail to produce memorable NPC moments at high rates), but a controlled comparison with matched-quality session designs would be required to establish causation cleanly.

6 Conclusion

NPC emotional moments in TRPG are often the sessions that players remember longest — and the hardest for adventure designers to produce reliably. The conventional approach is to design rich NPC backstories and trust the DM to deploy them at the right moment. This works when DMs are experienced, attuned to player state, and operating with low cognitive load. It fails — silently, without obvious error — when those conditions are not met.

The NPC arc-completion pattern addresses this failure mode by moving the essential design work from runtime to design time. The question "what should this NPC say, and when?" is answered before play begins: the arc-completion act is specified verbatim, the firing condition is specified as a concrete observable party action, and the DM discipline note specifies what must not happen after. This pre-specification does not constrain play; it enables the party's actions to become the cause of the NPC's most significant moment, which is a richer design than any DM-authored monologue can produce.

The Marathon corpus documents 52 instances of this pattern firing correctly across 49 sessions and 7 campaigns. Session coverage is 90%. Zero instances required DM scripting. The pattern holds for morally grey NPCs, professional-code NPCs, information-holders, craft-witnesses, and situationally-triggered arcs — not only in the grief-adjacent campaigns where it was first documented. The failure analysis confirms that NPCs without arc-completion conditions produce memorable moments in fewer than 10% of sessions where they are the primary NPC encounter, even in high-scoring sessions.

Three design principles are directly actionable from this evidence:

(1) **Specify the act and the condition before play.** Every significant NPC should have an arc-completion act (the specific thing they are capable of saying or doing) and a firing condition (the specific observable party action that triggers it) written into their design document before the first session. If you cannot specify these two things, you have not finished designing the NPC.
(2) **Make it smaller than you think it should be.** The arc-completion text should be the shortest formulation that is semantically complete. If it is longer than two sentences, it is probably not the arc-completion; it is the designer's wish for what should follow. The accumulated context of prior sessions does the work; the arc-completion needs only to be the hinge.
(3) **Enforce the discipline note.** Do not deliver the arc-completion before the firing condition is met. Do not extend it afterward. Do not soften it for morally grey NPCs. The discipline note is not a suggestion; it is the mechanism by which the party's actions become the cause of the moment rather than the DM's timing.

Future work should apply the pattern in human-run TRPG sessions and measure DM discipline compliance, player-experience outcomes, and transfer to systems with different NPC design conventions. The bilateral arc-completion sub-type — in which NPC and PC arc-completion conditions share a trigger domain and fire simultaneously — warrants its own study: the four bilateral instances in this corpus produced the highest single-scene quality scores observed, suggesting that the design of shared trigger domains between NPC and PC arcs is a high-leverage design intervention that has not yet been systematically investigated.

The claim this paper makes is modest and falsifiable: NPCs designed with explicit arc-completion conditions produce memorable moments more reliably than NPCs without them, and they do so without requiring the DM to script, time, or improvise the moment. The corpus evidence supports the claim. The design template is available, the discipline is learnable, and the failure mode — DM-scripted moments in place of character-logic moments — is identifiable and correctable. There is no excuse for NPCs that do nothing.

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